INTERNSHIP PROJECT

Enterprise Data Reconciliation

Turning messy transaction logs into clear financial insights

Python • Pandas • Regex • SQLite • Matplotlib

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Internship submission

Problem Statement

- Enterprise transaction data is often large, unstructured, and inconsistent, making accurate financial reconciliation difficult.

Key challenges

1. Large Dataset
   - Processing around 100,000 records efficiently requires optimized data operations.
2. Unstructured Logs
   - Transaction details must be extracted from messy server logs using Regex.
3. Missing Weekend Exchange Rates
   - Weekend transactions need appropriate exchange-rate matching to avoid losing records.
4. Historical Customer Records
   - The latest customer record must be identified before matching transactions.

Goal: Build a clean, accurate, and unified dataset for financial analysis.

Project Solution

- Python and Pandas: Process and clean large datasets efficiently.
- Regex: Extract transaction details from unstructured server logs.
- SQL Window Functions: Identify the latest customer records and filter active users.
- Exchange-rate matching: Convert EUR transactions into USD and handle weekend dates appropriately.
- Matplotlib / Seaborn: Visualize monthly revenue and the top five customers.

Expected outcome: A reconciled financial dataset and clear revenue insights.

What does the project do?

The goal is to combine messy business data and turn it into useful financial information.

MAIN OBJECTIVE

Create one reliable view of transactions, customers, and USD revenue.

The source data includes raw server logs, daily exchange rates, and customer history.

TOOLS USED

Python • Pandas • Regex • SQLite • Matplotlib / Seaborn

WORKFLOW

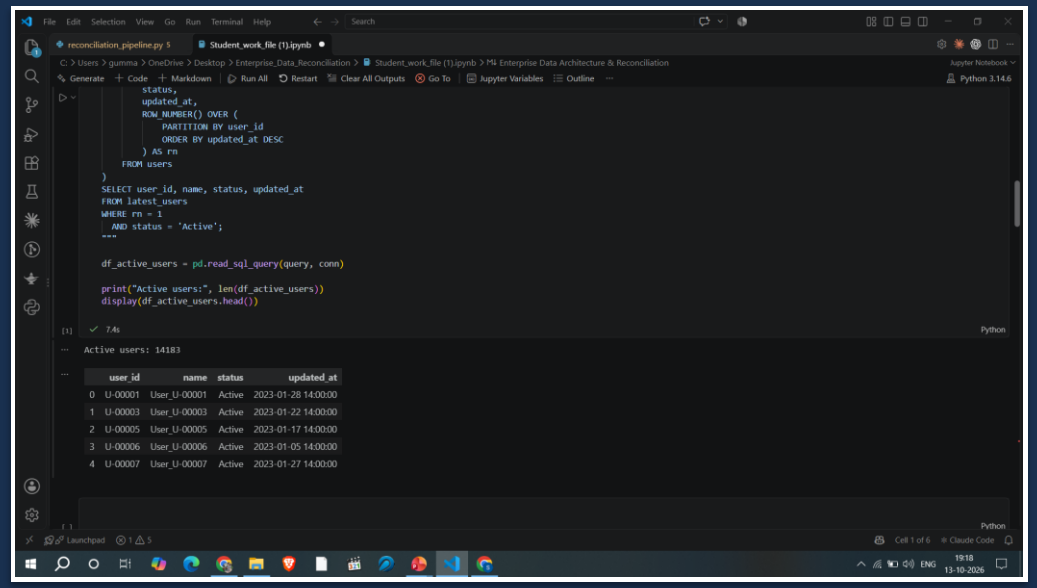
1. **Find latest customers**
   SQL window function
2. **Read and clean logs**
   Regex + Pandas
3. **Convert and reconcile**
   Exchange rates + joins
4. **Summarize results**
   Charts + tables

Result: a monthly revenue view and a top-customer summary.

Cleaning and extracting the source data

Screenshots from the notebook show the SQL and Regex steps.

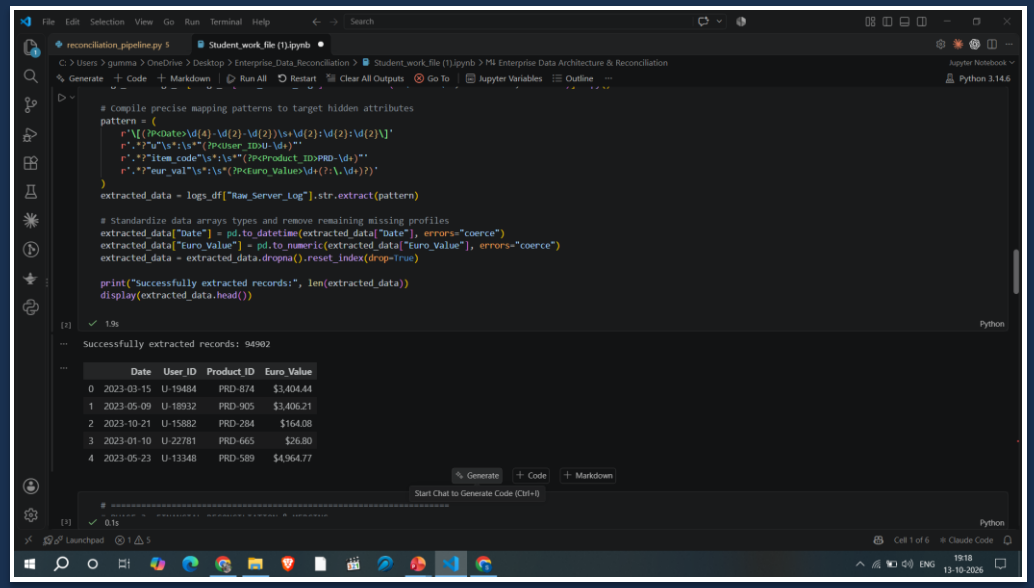
PHASE 1 • SQL CUSTOMER HISTORY



Selected the latest record per user and kept current Active users.

14,183 active users shown in output

PHASE 2 • REGEX LOG EXTRACTION

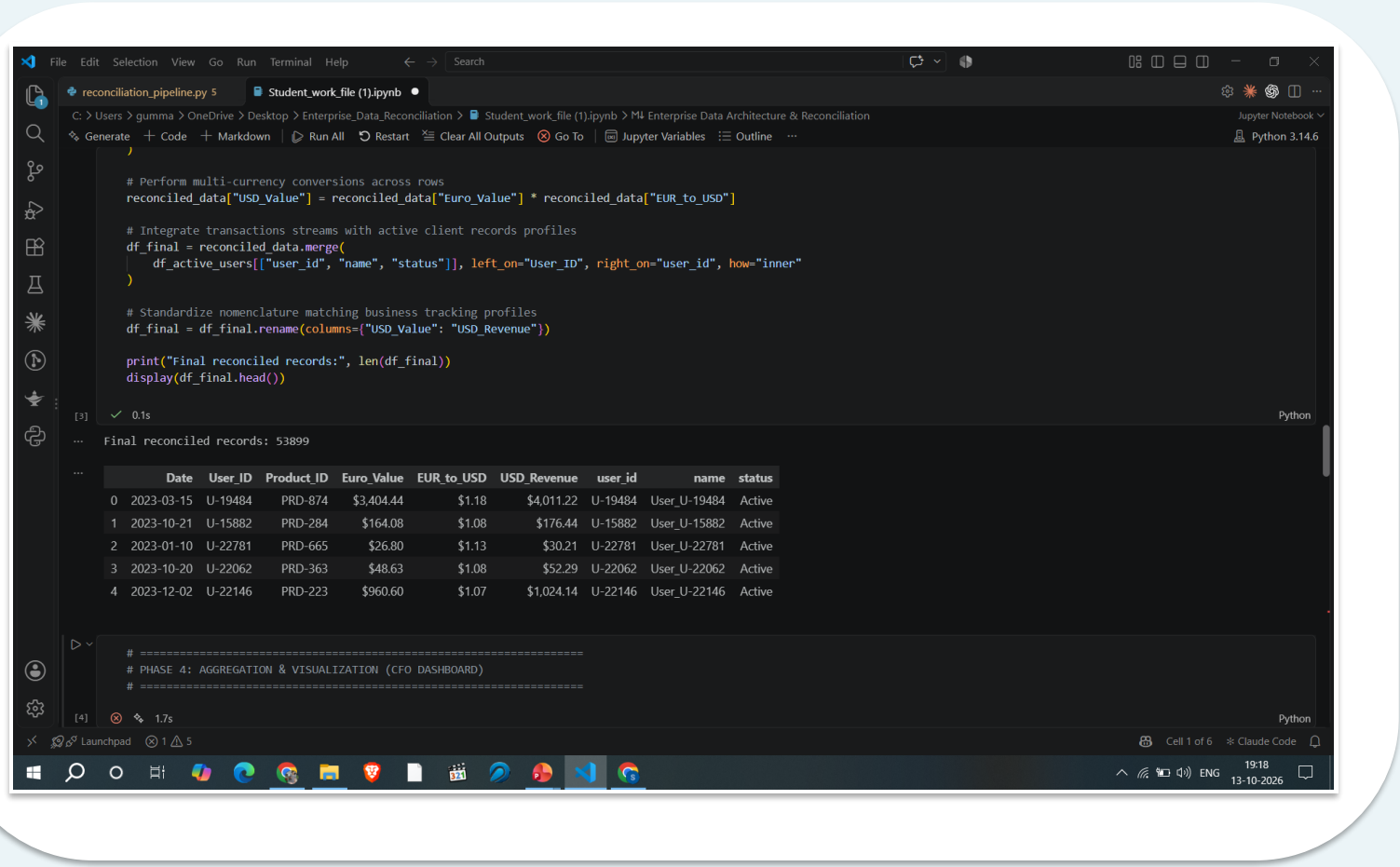


Extracted date, user, product, and euro amount from raw logs.

94,902 records extracted

Combining transactions, rates, and customers

The final table shows transaction values after currency conversion and customer matching.



KEY OUTPUT

53,899

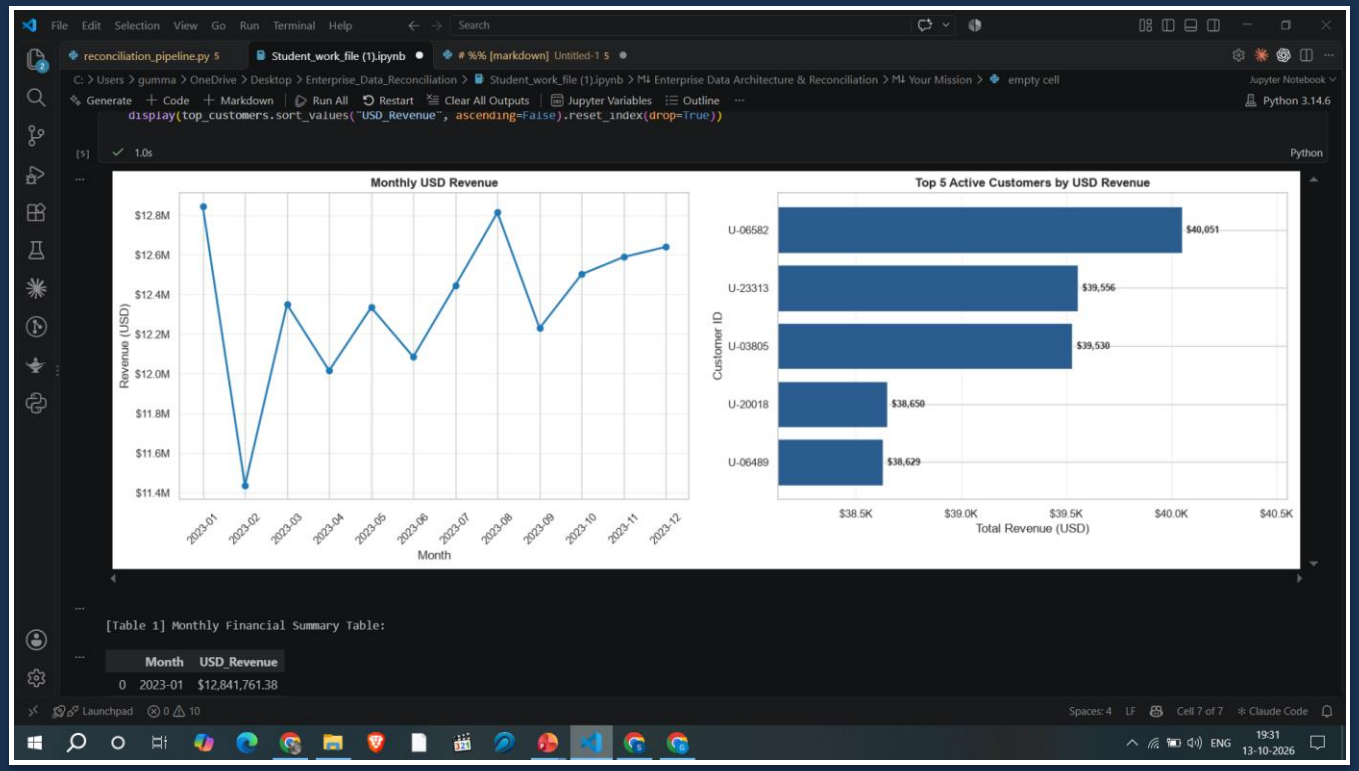
final reconciled records

EUR values were converted to USD and matched with active customer records.

Note: confirm weekend exchange-rate matching in the final run.

CFO dashboard: clear financial insights

The dashboard summarizes monthly revenue and highlights the top five customers.



Monthly revenue chart + top-five customer chart

KEY TAKEAWAYS

- Automated data extraction
- Combined transaction and customer data
- Created revenue summaries
- Visualized trends for review

Thank you